Evidence-as-Code Architecture Guide

DevOps.ai

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Contents

1 Evidence-as-Code Architecture Guide

1.1 Design Patterns and Implementation Strategies for CI/CD Evidence Automation

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1.2 Executive Summary

Evidence-as-Code treats compliance evidence as a first-class artifact in the software development lifecycle, automatically generating cryptographically signed, audit-ready evidence bundles from CI/CD pipelines. This approach eliminates manual evidence collection, reduces audit preparation time by 80%, and provides continuous compliance visibility.

This guide provides comprehensive design patterns, implementation strategies, and best practices for building Evidence-as-Code workflows using AlDeci, including crypto-

graphic signing, SLSA provenance, retention policies, and integration with compliance frameworks (SOC 2, ISO 27001, APRA CPS 234, Essential Eight).

Key Benefits: - Automate evidence collection from CI/CD pipelines - Generate cryptographically signed evidence bundles - Maintain immutable audit trails with 7-year retention - Reduce audit preparation time from weeks to hours

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1.4 1. Evidence-as-Code Principles

1.4.1 1.1 Core Principles

Principle 1: Evidence is Code - Evidence artifacts are version-controlled alongside application code - Evidence generation is automated in CI/CD pipelines - Evidence schemas are defined as code (JSON Schema, Protobuf)

Principle 2: Immutability - Evidence bundles are immutable once signed - Changes require new evidence bundles with provenance links - Audit trails are append-only and tamper-proof

Principle 3: Cryptographic Trust - All evidence is cryptographically signed - Signatures are verifiable by auditors - Public keys are published for transparency

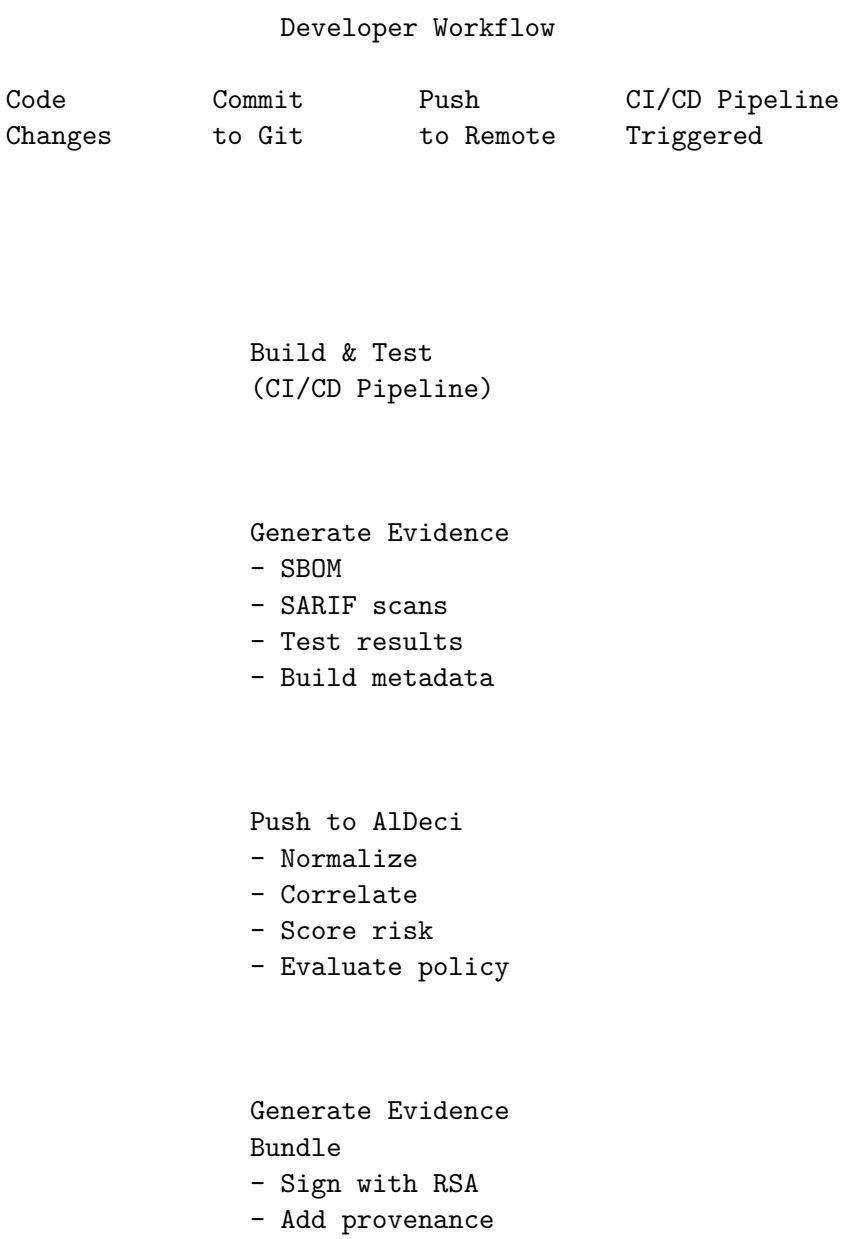
Principle 4: Continuous Compliance - Evidence is generated on every build/deployment - Compliance status is visible in real-time - Gaps are detected immediately, not at audit time

Principle 5: Explainability - Evidence includes clear rationale for decisions - Audit trails link evidence to requirements - Human-readable reports accompany machine-readable data

1.4.2 1.2 Benefits Over Manual Evidence Collection

Aspect	Manual Collection	Evidence-as-Code
Time	10-20 days per audit	2-4 hours per audit
Accuracy	Human error prone	Automated, consistent
Timeliness	Collected at audit time	Continuous generation
Completeness	May miss evidence	Comprehensive coverage
Verifiability	Unsigned documents	Cryptographically signed
Retention	Manual archiving	Automated 7-year retention
Cost	High labor cost	Low operational cost

1.4.3 1.3 Evidence-as-Code Workflow



```
- Store 7 years


Publish Evidence
- S3 storage
- Jira attachment
- Audit dashboard
```

1.5 2. Architecture Patterns

1.5.1 2.1 Pattern 1: Pipeline-Embedded Evidence Generation

Description: Evidence generation is embedded directly in CI/CD pipeline as a build step.

Use Case: Greenfield projects with full control over CI/CD pipelines.

Architecture:

```yaml
# .github/workflows/build.yml
name: Build and Generate Evidence

on: [push]

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3

      - name: Build application
        run: npm run build

      - name: Generate SBOM
        run: syft . -o cyclonedx-json > sbom.json

      - name: Run SAST
        run: semgrep --sarif > sarif.json

      - name: Run tests
        run: npm test -- --coverage

      - name: Push evidence to AlDeci
        run: |
          curl -X POST https://api.devops.ai/products/aldeci/inputs/sbom \
            -H "Authorization: Bearer ${{ secrets.ALDECI_TOKEN }}" \
```

```
          -F "file=@sbom.json"

        curl -X POST https://api.devops.ai/products/aldeci/inputs/sarif \
          -H "Authorization: Bearer ${{ secrets.ALDECI_TOKEN }}" \
          -F "file=@sarif.json"

    - name: Generate evidence bundle
      run: |
        curl -X POST https://api.devops.ai/products/aldeci/evidence/generate \
          -H "Authorization: Bearer ${{ secrets.ALDECI_TOKEN }}" \
          -H "Content-Type: application/json" \
          -d '{
            "component": "my-app",
            "version": "${{ github.sha }}",
            "compliance_frameworks": ["soc2", "iso27001"]
          }' \
          -o evidence-bundle.zip

    - name: Upload evidence bundle
      uses: actions/upload-artifact@v3
      with:
        name: evidence-bundle
        path: evidence-bundle.zip
        retention-days: 2555  # 7 years
```

Pros: - Simple to implement - Evidence generated on every build - No additional infrastructure required

Cons: - Couples evidence generation to build pipeline - May slow down build times - Requires pipeline modification for each project

1.5.2 2.2 Pattern 2: Sidecar Evidence Collector

Description: Evidence collection runs as a sidecar process that monitors CI/CD artifacts.

Use Case: Brownfield projects with existing CI/CD pipelines that can't be easily modified.

Architecture:

```
                    CI/CD Pipeline

  Build    →    Test     →    Scan     →    Deploy
```

```
Artifact Storage
(S3, Artifactory)
```

```
Evidence Collector
(Sidecar Process)
- Watch for artifacts
- Push to AlDeci
- Generate bundles
```

```
AlDeci
- Normalize
- Correlate
- Generate evidence
```

Implementation:

```python
# evidence_collector.py
import boto3
import requests
import time

s3 = boto3.client('s3')
aldeci_api = "https://api.devops.ai/products/aldeci"
aldeci_token = os.getenv("ALDECI_TOKEN")

def watch_artifacts():
    """Watch S3 bucket for new artifacts and push to AlDeci"""
    while True:
        # List new artifacts
        response = s3.list_objects_v2(
            Bucket='ci-artifacts',
            Prefix='builds/',
            StartAfter=last_processed_key
        )

        for obj in response.get('Contents', []):
            key = obj['Key']

            # Download artifact
```

```
            artifact = s3.get_object(Bucket='ci-artifacts', Key=key)

            # Determine artifact type
            if key.endswith('.sbom.json'):
                push_sbom(artifact['Body'].read())
            elif key.endswith('.sarif'):
                push_sarif(artifact['Body'].read())
            elif key.endswith('.test-results.json'):
                push_test_results(artifact['Body'].read())

            last_processed_key = key

        time.sleep(60)  # Check every minute

def push_sbom(sbom_data):
    """Push SBOM to AlDeci"""
    response = requests.post(
        f"{aldeci_api}/inputs/sbom",
        headers={"Authorization": f"Bearer {aldeci_token}"},
        files={"file": sbom_data}
    )
    response.raise_for_status()

def push_sarif(sarif_data):
    """Push SARIF to AlDeci"""
    response = requests.post(
        f"{aldeci_api}/inputs/sarif",
        headers={"Authorization": f"Bearer {aldeci_token}"},
        files={"file": sarif_data}
    )
    response.raise_for_status()
```

Pros: - No modification to existing pipelines - Centralized evidence collection - Can collect from multiple CI/CD systems

Cons: - Additional infrastructure required - Potential delay in evidence generation - Requires artifact storage integration

1.5.3 2.3 Pattern 3: GitOps Evidence Reconciliation

Description: Evidence generation triggered by GitOps reconciliation loops (ArgoCD, Flux).

Use Case: Kubernetes-native applications using GitOps deployment patterns.

Architecture:

```
                    Git Repository
```

```
manifests/
  deployment.yaml
  service.yaml
  evidence-policy.yaml  ← Evidence requirements
```

```
ArgoCD / Flux
- Sync manifests
- Deploy to K8s
- Trigger evidence
```

```
Evidence Controller
(K8s Operator)
- Watch deployments
- Generate SBOM
- Scan images
- Push to AlDeci
```

```
AlDeci
- Generate evidence
- Validate policy
- Store bundles
```

Implementation:

```yaml
# evidence-policy.yaml
apiVersion: evidence.devops.ai/v1
kind: EvidencePolicy
metadata:
  name: my-app-evidence
  namespace: production
spec:
  component: my-app
  version: v2.1.0
  complianceFrameworks:
    - soc2
    - iso27001
  evidenceRequirements:
```

```
    - type: sbom
      format: cyclonedx
      required: true
    - type: sarif
      scanTypes: [SAST, SCA, container]
      required: true
    - type: provenance
      format: slsa-v1
      required: true
  retentionDays: 2555  # 7 years
  signing:
    algorithm: RSA-SHA256
    keyRef: evidence-signing-key
```

Pros: - Native Kubernetes integration - Declarative evidence requirements - Automatic evidence on every deployment

Cons: - Requires Kubernetes - Additional operator deployment - More complex setup

---

1.6 3. Cryptographic Signing

1.6.1 3.1 Signing Algorithms

AIDeci supports two signing algorithms:

RSA-SHA256 (Recommended for most use cases) - Key size: 2048-bit or 4096-bit - Signature size: 256 bytes (2048-bit) or 512 bytes (4096-bit) - Performance: ~1ms per signature - Compatibility: Widely supported, auditor-friendly

Cosign (Recommended for container images) - Based on Sigstore project - Keyless signing with OIDC - Transparency log integration - Container-native

1.6.2 3.2 Key Management

Key Generation:

```
# Generate RSA key pair
openssl genrsa -out evidence-signing-key.pem 4096
openssl rsa -in evidence-signing-key.pem -pubout -out evidence-signing-key.pub

# Store private key securely
aws secretsmanager create-secret \
  --name aldeci-evidence-signing-key \
  --secret-string file://evidence-signing-key.pem

# Publish public key for verification
aws s3 cp evidence-signing-key.pub s3://public-keys/aldeci/evidence-signing-key.pub --acl publi
```

Key Rotation:

```
# Generate new key pair
openssl genrsa -out evidence-signing-key-2025.pem 4096
openssl rsa -in evidence-signing-key-2025.pem -pubout -out evidence-signing-key-2025.pub

# Update AlDeci configuration
curl -X PUT https://api.devops.ai/products/aldeci/config/signing \
  -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{
    "algorithm": "RSA-SHA256",
    "key_id": "2025-01",
    "key_path": "/app/keys/evidence-signing-key-2025.pem",
    "public_key_url": "https://public-keys.s3.amazonaws.com/aldeci/evidence-signing-key-2025.pu
  }'

# Old key remains valid for verification
# New evidence signed with new key
# Rotation complete
```

Key Storage Best Practices: - Store private keys in AWS Secrets Manager, Azure Key Vault, or HashiCorp Vault - Never commit private keys to Git - Use IAM roles for key access (no hardcoded credentials) - Enable key rotation every 12 months - Maintain old keys for verification of historical evidence

1.6.3 3.3 Signature Verification

Manual Verification:

```
# Download evidence bundle
wget https://evidence.devops.ai/bundles/evidence-20250126-abc123.zip

# Extract bundle
unzip evidence-20250126-abc123.zip

# Download public key
wget https://public-keys.s3.amazonaws.com/aldeci/evidence-signing-key.pub

# Verify signature
openssl dgst -sha256 -verify evidence-signing-key.pub \
  -signature signatures/manifest.sig \
  manifest.yaml

# Output: Verified OK
```

Automated Verification:

```
# verify_evidence.py
import hashlib
```

```python
from cryptography.hazmat.primitives import hashes, serialization
from cryptography.hazmat.primitives.asymmetric import padding
from cryptography.hazmat.backends import default_backend

def verify_evidence_bundle(bundle_path, public_key_url):
    """Verify evidence bundle signature"""

    # Extract bundle
    with zipfile.ZipFile(bundle_path, 'r') as zip_ref:
        zip_ref.extractall('evidence')

    # Read manifest
    with open('evidence/manifest.yaml', 'rb') as f:
        manifest_data = f.read()

    # Read signature
    with open('evidence/signatures/manifest.sig', 'rb') as f:
        signature = f.read()

    # Download public key
    response = requests.get(public_key_url)
    public_key = serialization.load_pem_public_key(
        response.content,
        backend=default_backend()
    )

    # Verify signature
    try:
        public_key.verify(
            signature,
            manifest_data,
            padding.PKCS1v15(),
            hashes.SHA256()
        )
        return True
    except Exception as e:
        print(f"Verification failed: {e}")
        return False

# Usage
if verify_evidence_bundle('evidence-20250126-abc123.zip', 'https://public-keys.s3.amazonaws.co
    print("Evidence bundle verified successfully")
else:
    print("Evidence bundle verification failed")
```

1.7 4. SLSA Provenance

1.7.1 4.1 SLSA Overview

SLSA (Supply-chain Levels for Software Artifacts) is a framework for ensuring software supply chain integrity. AlDeci generates SLSA v1 provenance attestations for all evidence bundles.

SLSA Levels: - **SLSA 1**: Provenance exists (build metadata documented) - **SLSA 2**: Provenance is signed (tamper-proof) - **SLSA 3**: Provenance is non-forgeable (isolated build environment) - **SLSA 4**: Provenance is auditable (two-party review)

AlDeci provides **SLSA Level 2** provenance by default (signed provenance).

1.7.2 4.2 Provenance Attestation Structure

```
{
  "_type": "https://in-toto.io/Statement/v0.1",
  "subject": [
    {
      "name": "evidence-20250126-abc123.zip",
      "digest": {
        "sha256": "a1b2c3d4e5f6..."
      }
    }
  ],
  "predicateType": "https://slsa.dev/provenance/v1",
  "predicate": {
    "buildDefinition": {
      "buildType": "https://devops.ai/aldeci/evidence-generation/v1",
      "externalParameters": {
        "component": "my-app",
        "version": "v2.1.0",
        "compliance_frameworks": ["soc2", "iso27001"]
      },
      "internalParameters": {
        "aldeci_version": "2.1.0",
        "evidence_policy": "default"
      },
      "resolvedDependencies": [
        {
          "uri": "git+https://github.com/example/my-app@abc123",
          "digest": {"sha1": "abc123def456..."}
        }
      ]
    },
    "runDetails": {
      "builder": {
        "id": "https://github.com/actions/runner/v2.311.0"
```

```
      },
      "metadata": {
        "invocationId": "https://github.com/example/my-app/actions/runs/123456",
        "startedOn": "2025-01-26T10:00:00Z",
        "finishedOn": "2025-01-26T10:30:00Z"
      },
      "byproducts": [
        {
          "name": "sbom.json",
          "digest": {"sha256": "f6e5d4c3b2a1..."}
        },
        {
          "name": "sarif.json",
          "digest": {"sha256": "1a2b3c4d5e6f..."}
        }
      ]
    }
  }
}
```

1.7.3 4.3 Provenance Generation

AIDeci automatically generates provenance attestations for all evidence bundles:

```python
# Automatic provenance generation
from services.provenance.attestation import generate_slsa_provenance

provenance = generate_slsa_provenance(
    subject_name="evidence-20250126-abc123.zip",
    subject_digest="a1b2c3d4e5f6...",
    builder_id="https://github.com/actions/runner/v2.311.0",
    source_uri="git+https://github.com/example/my-app@abc123",
    source_digest="abc123def456...",
    build_config={
        "component": "my-app",
        "version": "v2.1.0",
        "compliance_frameworks": ["soc2", "iso27001"]
    },
    materials=[
        {"uri": "sbom.json", "digest": "f6e5d4c3b2a1..."},
        {"uri": "sarif.json", "digest": "1a2b3c4d5e6f..."}
    ]
)

# Sign provenance
signed_provenance = sign_provenance(provenance, signing_key)
```

```
# Include in evidence bundle
bundle.add_file("provenance_attestation.json", signed_provenance)
```

1.8 5. Retention and Governance

1.8.1 5.1 Retention Policies

Regulatory Requirements: - SOC 2: 7 years - ISO 27001: 3 years (recommended 7 years) - APRA CPS 234: 7 years - GDPR: 6 years (UK), varies by jurisdiction - HIPAA: 6 years

AIDeci Default: 7 years (2555 days)

Configuration:

```
# config/evidence.yml
evidence:
  retention_days: 2555  # 7 years
  storage:
    backend: s3
    bucket: evidence-bundles
    region: ap-southeast-2
    encryption: AES-256
    lifecycle_policy: glacier_after_90_days
  cleanup:
    enabled: true
    schedule: "0 0 * * 0"  # Weekly on Sunday
    dry_run: false
```

1.8.2 5.2 Storage Architecture

Tiered Storage:

```
                    Evidence Lifecycle

  Hot           Warm          Cold          Archive
  (0-90 days)  (90-365 days  (1-7 years)   (> 7 years)
  S3 Standard  S3 IA         S3 Glacier    Delete
  $0.023/GB    $0.0125/GB    $0.004/GB     -
```

S3 Lifecycle Policy:

```
{
  "Rules": [
    {
      "Id": "evidence-lifecycle",
```

```
      "Status": "Enabled",
      "Transitions": [
        {
          "Days": 90,
          "StorageClass": "STANDARD_IA"
        },
        {
          "Days": 365,
          "StorageClass": "GLACIER"
        }
      ],
      "Expiration": {
        "Days": 2555
      }
    }
  ]
}
```

1.8.3 5.3 Access Control

IAM Policy (AWS):

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AlDeciEvidenceWrite",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::123456789012:role/aldeci-evidence-writer"
      },
      "Action": [
        "s3:PutObject",
        "s3:PutObjectAcl"
      ],
      "Resource": "arn:aws:s3:::evidence-bundles/*"
    },
    {
      "Sid": "AuditorEvidenceRead",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::123456789012:role/auditor"
      },
      "Action": [
        "s3:GetObject",
        "s3:ListBucket"
      ],
```

```
      "Resource": [
        "arn:aws:s3:::evidence-bundles",
        "arn:aws:s3:::evidence-bundles/*"
      ]
    },
    {
      "Sid": "DenyDelete",
      "Effect": "Deny",
      "Principal": "*",
      "Action": [
        "s3:DeleteObject",
        "s3:DeleteObjectVersion"
      ],
      "Resource": "arn:aws:s3:::evidence-bundles/*"
    }
  ]
}
```

1.9 6. CI/CD Integration

1.9.1 6.1 GitHub Actions

```yaml
# .github/workflows/evidence.yml
name: Generate Evidence

on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

jobs:
  evidence:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3

      - name: Generate SBOM
        run: |
          curl -sSfL https://raw.githubusercontent.com/anchore/syft/main/install.sh | sh -s --
          syft . -o cyclonedx-json > sbom.json

      - name: Run SAST
        run: |
          pip install semgrep
```

```
        semgrep --config=auto --sarif > sarif.json

    - name: Push to AlDeci
      env:
        ALDECI_TOKEN: ${{ secrets.ALDECI_TOKEN }}
      run: |
        curl -X POST https://api.devops.ai/products/aldeci/inputs/sbom \
          -H "Authorization: Bearer $ALDECI_TOKEN" \
          -F "file=@sbom.json" \
          -F "component=${{ github.repository }}" \
          -F "version=${{ github.sha }}"

        curl -X POST https://api.devops.ai/products/aldeci/inputs/sarif \
          -H "Authorization: Bearer $ALDECI_TOKEN" \
          -F "file=@sarif.json" \
          -F "component=${{ github.repository }}" \
          -F "version=${{ github.sha }}"

    - name: Generate evidence bundle
      env:
        ALDECI_TOKEN: ${{ secrets.ALDECI_TOKEN }}
      run: |
        curl -X POST https://api.devops.ai/products/aldeci/evidence/generate \
          -H "Authorization: Bearer $ALDECI_TOKEN" \
          -H "Content-Type: application/json" \
          -d '{
            "component": "${{ github.repository }}",
            "version": "${{ github.sha }}",
            "compliance_frameworks": ["soc2", "iso27001"],
            "metadata": {
              "build_url": "${{ github.server_url }}/${{ github.repository }}/actions/runs/$
              "commit_message": "${{ github.event.head_commit.message }}",
              "author": "${{ github.actor }}"
            }
          }' \
          -o evidence-bundle.zip

    - name: Upload evidence bundle
      uses: actions/upload-artifact@v3
      with:
        name: evidence-bundle
        path: evidence-bundle.zip
```

1.9.2 6.2 GitLab CI

```yaml
# .gitlab-ci.yml
stages:
  - build
  - scan
  - evidence

build:
  stage: build
  script:
    - npm run build
  artifacts:
    paths:
      - dist/

scan:
  stage: scan
  script:
    - syft . -o cyclonedx-json > sbom.json
    - semgrep --config=auto --sarif > sarif.json
  artifacts:
    paths:
      - sbom.json
      - sarif.json

evidence:
  stage: evidence
  script:
    - |
      curl -X POST https://api.devops.ai/products/aldeci/inputs/sbom \
        -H "Authorization: Bearer $ALDECI_TOKEN" \
        -F "file=@sbom.json" \
        -F "component=$CI_PROJECT_PATH" \
        -F "version=$CI_COMMIT_SHA"

    - |
      curl -X POST https://api.devops.ai/products/aldeci/inputs/sarif \
        -H "Authorization: Bearer $ALDECI_TOKEN" \
        -F "file=@sarif.json" \
        -F "component=$CI_PROJECT_PATH" \
        -F "version=$CI_COMMIT_SHA"

    - |
      curl -X POST https://api.devops.ai/products/aldeci/evidence/generate \
        -H "Authorization: Bearer $ALDECI_TOKEN" \
        -H "Content-Type: application/json" \
        -d "{
          \"component\": \"$CI_PROJECT_PATH\",
```

```
        \"version\": \"$CI_COMMIT_SHA\",
        \"compliance_frameworks\": [\"soc2\", \"iso27001\"],
        \"metadata\": {
          \"build_url\": \"$CI_PIPELINE_URL\",
          \"commit_message\": \"$CI_COMMIT_MESSAGE\",
          \"author\": \"$GITLAB_USER_EMAIL\"
        }
      }" \
      -o evidence-bundle.zip
artifacts:
  paths:
    - evidence-bundle.zip
```

1.10 7. Compliance Framework Mapping

1.10.1 7.1 SOC 2 Type II

Control	Evidence Type	AIDeci Artifact
CC6.1 - Logical access controls	Access logs	IAM policies, RBAC config
CC6.6 - Vulnerability management	Scan results	SARIF findings, risk reports
CC7.1 - Change management	Build metadata	Provenance attestations
CC7.2 - System monitoring	Observability	Metrics, traces, logs
CC8.1 - Risk assessment	Risk scores	FixOpsRisk reports

1.10.2 7.2 ISO 27001

Control	Evidence Type	AIDeci Artifact
A.12.6.1 - Vulnerability management	Scan results	SARIF findings, risk reports
A.14.2.1 - Secure development policy	SAST results	SARIF findings (SAST)
A.14.2.5 - Secure system engineering	SBOM	Normalized SBOM, dependencies
A.18.2.3 - Technical compliance review	Compliance reports	Policy evaluations

1.10.3 7.3 APRA CPS 234

Requirement	Evidence Type	AIDeci Artifact
Req 17 - Vulnerability management	Scan results	SARIF findings, risk reports
Req 18 - Patch management	Version lag	SBOM with version lag analysis
Req 28 - Control testing	Test results	Scan frequency, coverage metrics
Req 31 - Vulnerability remediation	Remediation tracking	MTTR metrics, Jira tickets

---

1.11 8. Implementation Examples

1.11.1 8.1 Example: Microservices Architecture

Scenario: 50 microservices, each with independent CI/CD pipelines

Solution: Centralized evidence collection with per-service evidence bundles

```yaml
# evidence-config.yml (per service)
service:
  name: user-service
  version: v2.1.0
  compliance_frameworks:
    - soc2
    - iso27001
  evidence_requirements:
    - type: sbom
      format: cyclonedx
      frequency: every_build
    - type: sarif
      scan_types: [SAST, SCA]
      frequency: every_build
    - type: container_scan
      scan_types: [vulnerability, malware]
      frequency: every_build
  retention_days: 2555
```

CI/CD Integration:

```bash
# Each service pipeline
syft . -o cyclonedx-json > sbom.json
semgrep --config=auto --sarif > sarif.json
trivy image --format sarif user-service:v2.1.0 > container-scan.sarif

# Push to AlDeci
aldeci push sbom sbom.json --service user-service --version v2.1.0
aldeci push sarif sarif.json --service user-service --version v2.1.0
```

```
aldeci push sarif container-scan.sarif --service user-service --version v2.1.0

# Generate evidence bundle
aldeci evidence generate --service user-service --version v2.1.0 --output evidence-bundle.zip
```

1.11.2 8.2 Example: Monorepo

Scenario: Single monorepo with multiple applications

Solution: Component-level evidence generation with shared compliance configuration

```
# .aldeci/config.yml (monorepo root)
monorepo:
  name: my-monorepo
  components:
    - name: frontend
      path: apps/frontend
      compliance_frameworks: [soc2]
    - name: backend
      path: apps/backend
      compliance_frameworks: [soc2, iso27001]
    - name: worker
      path: apps/worker
      compliance_frameworks: [soc2]
  shared_evidence:
    - type: repository_scan
      frequency: daily
    - type: dependency_scan
      frequency: weekly
```

CI/CD Integration:

```
# Generate evidence for each component
for component in frontend backend worker; do
  syft apps/$component -o cyclonedx-json > sbom-$component.json
  semgrep --config=auto apps/$component --sarif > sarif-$component.json

  aldeci push sbom sbom-$component.json --component $component --version $COMMIT_SHA
  aldeci push sarif sarif-$component.json --component $component --version $COMMIT_SHA
done

# Generate monorepo-level evidence
aldeci evidence generate-monorepo --config .aldeci/config.yml --version $COMMIT_SHA
```

1.12 9. Best Practices

1.12.1 9.1 Evidence Generation

1. **Generate evidence on every build** - Don't wait for audits
2. **Include build metadata** - Commit SHA, build URL, author
3. **Sign immediately** - Don't delay signing
4. **Validate before storing** - Check evidence completeness
5. **Use consistent naming** - `evidence-YYYYMMDD-{id}.zip`

1.12.2 9.2 Key Management

1. **Rotate keys annually** - Maintain old keys for verification
2. **Use hardware security modules (HSM)** - For production keys
3. **Publish public keys** - Make verification easy for auditors
4. **Separate signing keys** - Different keys for different purposes
5. **Audit key usage** - Log all signing operations

1.12.3 9.3 Storage and Retention

1. **Use tiered storage** - Hot/warm/cold based on age
2. **Enable versioning** - Protect against accidental deletion
3. **Replicate across regions** - Disaster recovery
4. **Encrypt at rest** - AES-256 or better
5. **Automate lifecycle** - Don't rely on manual cleanup

1.12.4 9.4 Compliance Mapping

1. **Document control mappings** - Link evidence to requirements
2. **Maintain evidence matrix** - Track coverage gaps
3. **Update mappings regularly** - As frameworks evolve
4. **Include rationale** - Explain why evidence satisfies control
5. **Get auditor buy-in** - Validate approach before audit

1.13 10. Troubleshooting

1.13.1 10.1 Common Issues

Issue: Evidence bundle generation fails with "missing SBOM"

Solution:

```
# Verify SBOM was pushed
curl -X GET https://api.devops.ai/products/aldeci/inputs/sbom/list \
  -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"component": "my-app", "version": "v2.1.0"}'
```

```
# If missing, push SBOM
curl -X POST https://api.devops.ai/products/aldeci/inputs/sbom \
  -H "Authorization: Bearer $TOKEN" \
  -F "file=@sbom.json" \
  -F "component=my-app" \
  -F "version=v2.1.0"
```

Issue: Signature verification fails

Solution:

```
# Check public key URL
curl -I https://public-keys.s3.amazonaws.com/aldeci/evidence-signing-key.pub

# Download correct public key
wget https://public-keys.s3.amazonaws.com/aldeci/evidence-signing-key.pub

# Verify with correct key
openssl dgst -sha256 -verify evidence-signing-key.pub \
  -signature signatures/manifest.sig \
  manifest.yaml
```

Issue: Evidence bundle too large (> 100 MB)

Solution:

```
# Exclude large files from evidence bundle
evidence:
  bundle_contents:
    - normalized_sbom
    - risk_report
    - sarif_findings  # Exclude raw SARIF (can be large)
    - provenance_attestation
    - policy_evaluations
    - manifest
  compression: gzip  # Enable compression
  max_size_mb: 100
```

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